

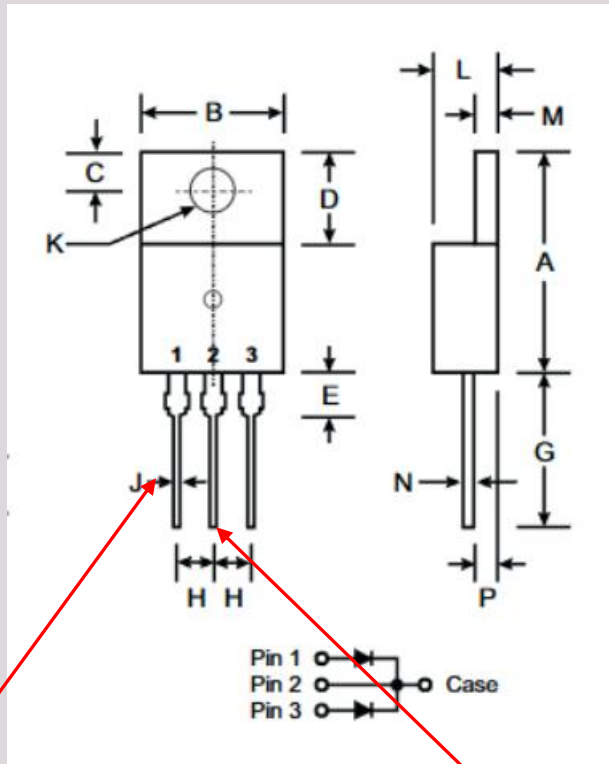
Prototype PCB Design

Week 11 — PV Integrated EV System

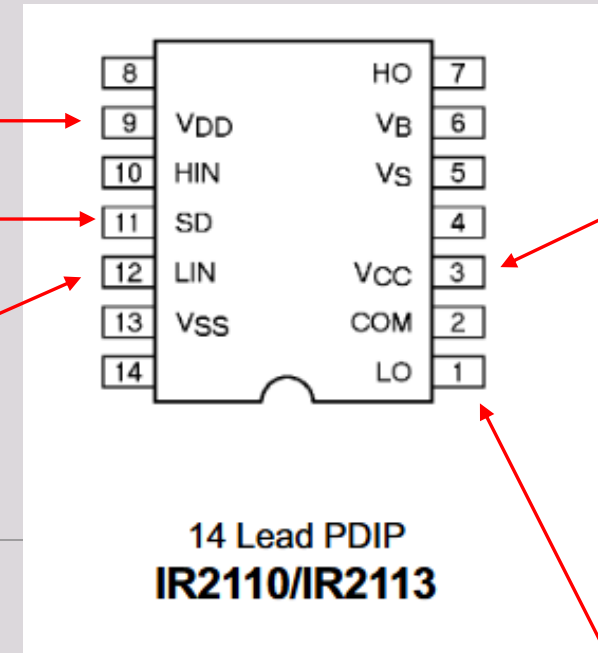
Component List

- PV Panel — 17.93V, 20W
- Inductor (L) — 2200 μ H, hand-wound toroidal
- Diode — MBR10100CT (Schottky)
- Capacitor (C) — 33 μ F / 100V
- MOSFET — IRF540N
- Load resistor — 120 Ω / 25W
- Gate driver — IR2110
- Microcontroller — Arduino Uno
- Gate resistor — 10 Ω
- Gate pull-down resistor — 10k Ω
- R1 (feedback divider) — 100k Ω
- R2 (feedback divider) — 10k Ω

MBR10100CT Diode



IR2110 Gatedriver



Pin 9 → +5v of Arduino

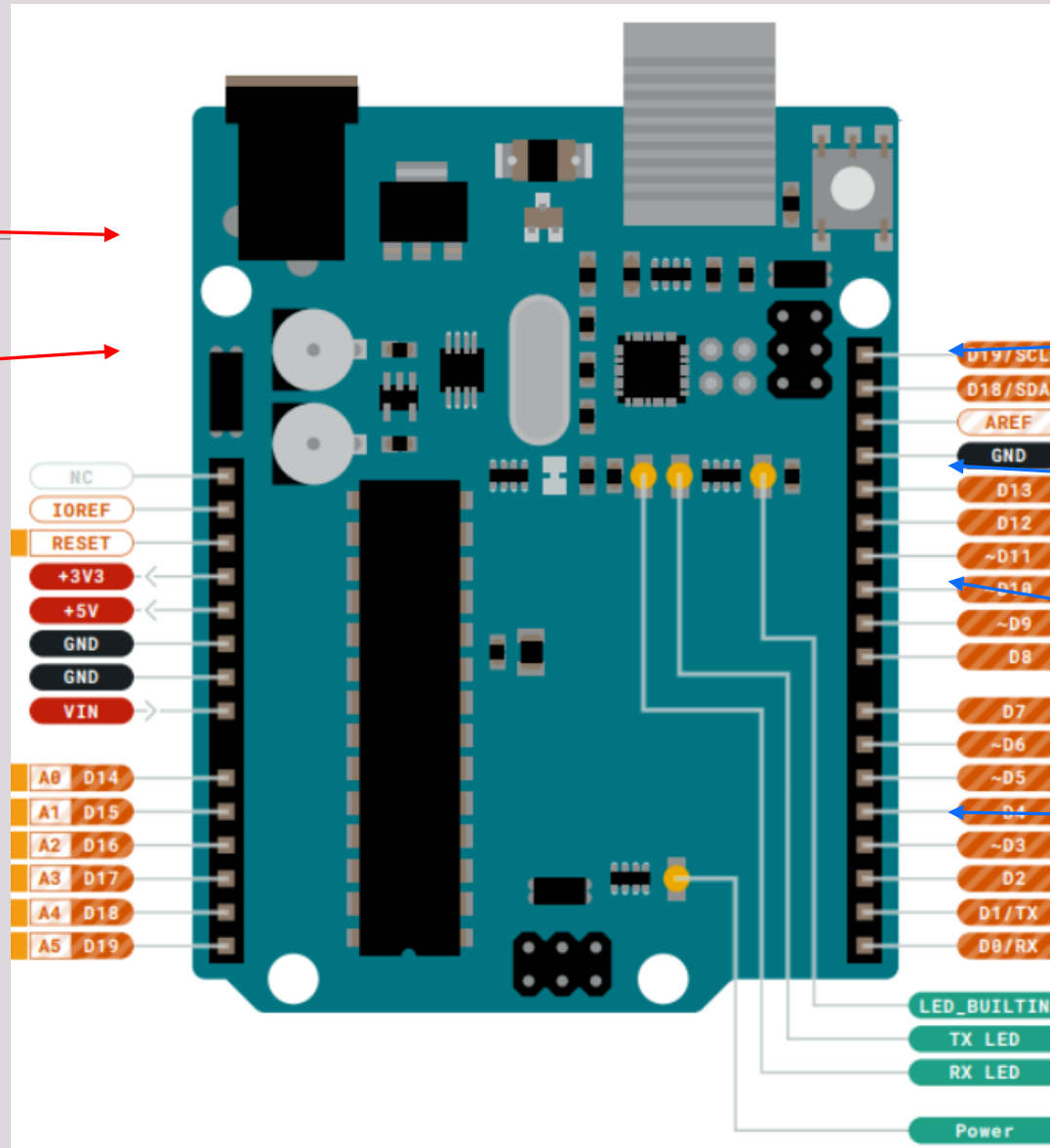
Pin 12 → D9 Arduino

Pin 3 → SOLAR PANEL + TERMINAL

Pin 1 → Gate of MOSFET

Pin 1 (LO) → R4 → Q2 Gate

Pin 2 (COM) → Ground /
MOSFET Source



Pin 13 (VSS) → Ground

Pin 12 (LIN) → Arduino digital pin
(PWM control input)

Pin 11 (SD) → Arduino digital pin
(shutdown control)

Pin 9 (VDD) → +5V supply
(from Arduino)

